

CPE 381 CLASSWORK 03: MATHEMATICAL PRELIMINARIES

Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes

Due: August 31, 2026 (In-Class)

1. Solve the differential equation $ty' + 3y = 0$, given $y(1) = 2$, assuming $t > 0$. Here $y' = \frac{dy}{dt}$. **Hint 1:** you will need to rearrange terms so that derivative term is on the left hand side and non-derivative terms on the right hand side, and then integrate. **Hint 2:** $\int \frac{1}{t} dt = \ln(t) + C$. **(5 Points)**

Answer

Rearranging the given differential equation,

$$y' + 3\frac{y}{t} = 0 \quad (1)$$

$$\frac{dy}{dt} + 3\frac{y}{t} = 0 \quad (2)$$

$$\Rightarrow \frac{dy}{dt} = -\frac{3y}{t} \quad (3)$$

$$\Rightarrow \frac{dy}{y} = -3\frac{dt}{t} \quad (4)$$

$$\Rightarrow \int \frac{dy}{y} = -3 \int \frac{dt}{t} \quad (5)$$

$$\Rightarrow \ln y = -3 \ln |t| + C, \quad C \text{ is the constant of integration} \quad (6)$$

$$\Rightarrow y = e^{-3 \ln |t| + C} = e^{-3 \ln |t|} e^C \quad (7)$$

Let $A = e^C$, then

$$\Rightarrow y = A \cdot e^{-3 \ln |t|}$$

$$y = At^{-3}$$

$$\text{As } y(1) = 2 \Rightarrow 2 = A \cdot (1)^{-3} \Rightarrow A = 2$$

$$\text{Thus, } y = 2t^{-3}$$

2. Find the derivate of $f(x) = x^2$ using the first priciple of the derivate. **(5 Points)**

Answer

From the first principle of the derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (8)$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \quad (9)$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + h^2 + 2hx - x^2}{h} \quad (10)$$

$$= \lim_{h \rightarrow 0} \frac{h(h+2x)}{h} \quad (11)$$

$$= 2x \quad (12)$$

3. Find the second, fourth, seventh, and ninth terms of the geometric progression with first term $a = 2$ and ratio $r = 3$. Does the sum of this geometric progression converge? **(4 + 1 Points)**

Answer

Nth term of a geometric progression can be written

$$a_n = ar^{n-1}. \text{ So } a_2 = 2 \cdot 3^{2-1} = 6, a_4 = 2 \cdot 3^{4-1} = 54, a_7 = 2 \cdot 3^{7-1} = 1458, a_9 = 2 \cdot 3^{9-1} = 13122.$$

As $|r| \not< 1$, the sum of the geometric progression doesn't converge.

4. Find area under the curve for the shaded region in Figure 1. **Hint:** $\int x^n dx = \frac{1}{n+1} x^{n+1}$. **(5 Points)**

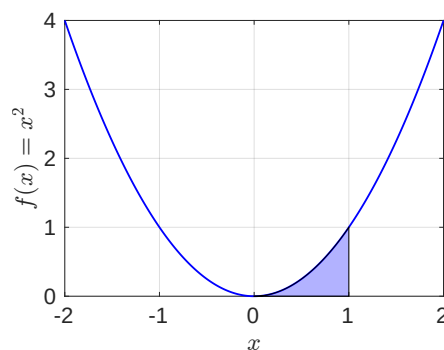


Figure 1: Area under curve

Answer

Area under the curve:

$$\int_0^1 x^2 dx = \left[\frac{1}{3} x^3 \right]_{x=0}^{x=1} = \frac{1}{3} \quad (13)$$